

Bilal Amin

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MSc Data Science and AI student at Saarland University, with current HiWi roles in software engineering research and web development. Interested in AI-assisted software engineering, HCI, UI/UX design, and web development.

EDUCATION

Universität des Saarlandes <i>Master's degree, Data Science and AI</i> <i>Grade: 1.9 (German grading system)</i>	Apr. 2024 – Present Saarbrücken, Germany
University of Bergen (Erasmus+ Exchange Semester) <i>Exchange Student, Computer Science</i>	Jan. 2026 – Jun. 2026 Bergen, Norway
National University of Sciences & Technology <i>Bachelor's in Mechatronics Engineering (Computer Science)</i> <i>Grade: 1.7 (German grading system)</i>	Sep. 2019 – Sep. 2023 Islamabad, Pakistan

EXPERIENCE

Research Assistant (HiWi) – Software Engineering <i>Chair of Software Engineering, Saarland University</i> - Work with Qingqing Dong on AI-assisted software engineering and model-driven systems engineering research. - Research focus includes model merging, semantic merge-conflict detection, and feature-based merge recommendations. - Support research on software evolution and intelligent methods for software and model-based systems.	Oct. 2026 – Present Saarbrücken, Germany
Student Assistant (HiWi) – Web Development <i>EuroComCenter, Saarland University</i> - Maintain and improve the EuroComCenter web platform, focusing on usability, multilingual content, and website structure. - Review the existing implementation, fix technical and user-interface issues, and manage CMS-based content and digital media. - Work with the project team to translate requirements into practical technical and design improvements.	Sep. 2026 – Present Saarbrücken, Germany
Working Student – Internet Architecture Group <i>Max Planck Institute for Informatics (MPI)</i> - Designed an automated framework to evaluate congestion control algorithms in NS-3 (Network Simulator). - Created a user-friendly dashboard to visualize performance metrics under different network conditions. - Developed system tests and metrics for performance evaluation across varied scenarios. - Analyzed simulation results to identify optimization opportunities and guide design decisions.	Oct. 2024 – Mar. 2025 Saarbrücken, Germany
Data Analyst <i>Convergent Business Technologies</i> - Developed reproducible data science workflows using Python and SQL to manipulate and analyze data. - As part of training, completed the CS50 course and created several dashboards related to the coursework.	Oct. 2023 – Mar. 2024 Islamabad, Pakistan
Web Development Intern <i>CureMD</i> - Designed user interfaces using HTML, CSS, and JavaScript, improving usability and accessibility for end users. - Developed APIs in the .NET framework, enhancing data integration and tool development.	Jun. 2023 – Aug. 2023 Lahore, Pakistan
Technical Intern <i>STEP (Space Technology Education and Popularization)</i> - Led workshops teaching programming fundamentals using Scratch, focusing on hands-on learning for underprivileged youth. - Created educational materials to simplify complex technology concepts, fostering interest in STEM fields.	Aug. 2021 – Oct. 2021 Islamabad, Pakistan

PROJECTS

AI-Assisted Dafny Code Generation and Verification

2026

- Built a Python/Codex pipeline that generates Dafny from English requirements and uses verifier feedback for up to two repairs.
- Evaluated Bounded Counter, Authorized Access, and Account Transfer with two prompt profiles: four of six runs verified initially; all six verified within three attempts. The profiles tied in this small experiment.
- Separately reviewed requirement coverage; formal verification checks written contracts rather than English-to-specification faithfulness.

GermanLyft – AI-Powered German Learning Platform

2026

- Developed a full-stack German-learning platform using React, TypeScript, Flask, and the Groq LLM API, mainly for CEFR A1–A2 learners.
- Built an interactive AI tutor for grammar explanations, sentence correction, vocabulary practice, conversational learning, and dynamically generated exercises.
- Implemented placement tests, quizzes, user accounts, profiles, and Anki .apkg deck importing using a REST-based React/Flask architecture.
- Deployed the frontend on Vercel and the backend on Render.

AI Learning Assistant using LLMs (RAG)

2025

- Built an AI-powered learning assistant using Python, LangChain, Streamlit, and FastAPI.
- Integrated a locally hosted Llama model with Retrieval-Augmented Generation to answer questions from uploaded PDF documents.
- Implemented document processing, vector embeddings, ChromaDB, and citation-based responses.
- Added features to generate summaries, quizzes, and study materials from course documents.

Smart StudyDesk: AI-Powered Study Organizer

2024

- Designed a cross-platform study application for organizing, summarizing, and reviewing lecture materials.
- Added AI-based note summaries, personalized quizzes, progress tracking, and weak-area analysis.
- Conducted usability testing using semi-structured interviews and NASA-TLX workload analysis.
- Improved the interface based on user feedback, with clearer navigation, folder organization, and feedback visibility.

MarketPulse Website Design and HCI Analysis

2024

- Designed and developed a responsive website for MarketPulse using HTML, CSS, and JavaScript.
- Applied HCI principles including consistency, simplicity, feedback, visibility, constraints, and accessibility.
- Improved the information structure and user interaction through usability-focused design.

Side Hustle Launchpad – Tech4Europe Hackathon, KIT

2024

- Developed a full-stack web application during the Tech4Europe Hackathon at Karlsruhe Institute of Technology, Germany.
- Built analytics dashboards and personalized recommendations to support small-business planning.
- Used FastAPI and MongoDB for backend development and data management.

MOOCs AND RELEVANT COURSEWORK

- Saarland University: Data Networks, Neural Networks: Theory and Implementation, Elements of Machine Learning, Human Computer Interaction, High-Level Computer Vision, Generative AI, Machine Translation, Attacks Against Machine Learning Models, and Competitive Programming.
- University of Bergen: Functional Programming, Qualitative Analysis, and Philosophy of AI.
- Certifications: Machine Learning Specialization – DeepLearning.AI and Stanford Online; CS50's Introduction to Artificial Intelligence with Python – Harvard University; Google Data Analytics Professional Certificate – Google; Python for Data Science, AI & Development – IBM.

TECHNICAL SKILLS AND INTERESTS

Programming: Python, JavaScript, TypeScript, SQL, HTML, CSS, C++, C#, Java, Dafny

Machine Learning & AI: Scikit-Learn, PyTorch, TensorFlow, Hugging Face Transformers, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG)

LLM Frameworks: LangChain, LangGraph, Ollama, ChromaDB, Codex SDK

Web Frameworks: FastAPI, Flask, Streamlit, ReactJS

Libraries & Data: Pandas, NumPy, PySpark, Matplotlib

Tools & Platforms: Git, GitHub Actions, Docker, Microsoft Azure, VS Code, Jupyter, Power BI, Tableau, Figma

Interests: LLM Applications, Agentic AI, AI-Assisted Software Engineering, Data Engineering, Workflow Automation, HCI, UI/UX Design